

INDRAJITH MP

Thrissur, Kerala 680662 | +91-7558865722 | indrajithmp77@gmail.com | linkedin.com/in/indrajithmp | github.com/Indra-jith | indrajithmp.vercel.app

TECHNICAL SKILLS

AI/ML & GenAI: Python, PyTorch, TensorFlow, Keras, Scikit-learn, Large Language Models (LLMs), Generative AI, Prompt Engineering, LLM Orchestration, Retrieval-Augmented Generation (RAG), Agentic AI (LangGraph, LangChain), Hugging Face, Embeddings, Vector Databases, Stable Diffusion
Data & Retrieval: Qdrant, Cohere Embeddings, Semantic Search, Vector Search, Chunking Strategies, Data Pipelines, Feature Engineering, EDA, Statistical Modeling, Pandas, NumPy
Computer Vision & Robotics: OpenCV, MediaPipe, Object Detection, Image Classification, Anomaly Detection, ROS2, Sensor Fusion (IMU, Ultrasonic, Encoders), Autonomous Navigation, Raspberry Pi Pico, Arduino, I2C, UART
Deployment & Engineering: FastAPI, Flask, REST APIs, WebSockets, Pydantic, BeautifulSoup, CI/CD, GitHub Actions, Cloudflare Workers, Model Serving, Render, Docker, C/C++, TypeScript, JavaScript

EXPERIENCE

- Moonpreneur (Silicon Valley-based Startup)

Feb 2025 – Present

Robotics & AI/ML Project Guide Intern

Remote

 - Cut autonomous navigation error rate by **40%** by engineering **4 sensor-fused robotic systems** on Raspberry Pi Pico, integrating ultrasonic, IMU, and encoder data via I2C/PWM with deterministic fusion algorithms.
 - Maintained **94% student comprehension rate** across **15+ live sessions** (50+ students) on neural networks, deep learning, computer vision, and embedded AI — verified via post-session assessments.
 - Drove **100% project completion** across 12 students through a 6-module ML curriculum spanning full end-to-end pipelines: data ingestion → feature engineering → model training → hyperparameter tuning → deployment.
 - Eliminated GPU dependency by shipping OpenCV + MediaPipe computer vision labs achieving **real-time object detection and hand tracking at 30 FPS** on commodity hardware via optimized inference pipelines.
- Allsoft Solutions (IBM Business Partner)

June 2024 – July 2024

AI/ML Intern

Remote

 - Delivered **97% test accuracy** and **0.96 F1-score** on a Fake News Detection classifier — ranked #1 in project scope — by training an ensemble (Logistic Regression, SVM, XGBoost, Naive Bayes) with GridSearchCV on **50K+ samples**.
 - Reduced label noise by **99.2%** via a statistical NLP preprocessing pipeline (tokenization, TF-IDF vectorization, stopword removal, exploratory data analysis) across LIAR dataset and Indian news corpora.

PROJECTS

- ShadowMind – Autonomous AI Investigation Agent | LangGraph, RAG, Qdrant, Cohere, FastAPI

2026 [In Progress]

 - Engineered a **hypothesis-driven agentic RAG pipeline** using LangGraph where the agent autonomously generates competing theories, retrieves targeted evidence per hypothesis from a **Qdrant vector database**, scores confidence, eliminates weak hypotheses, and loops until convergence — going beyond standard single-pass RAG.
 - Built a **custom web ingestion pipeline** using BeautifulSoup to scrape, clean, and chunk **150+ articles across 30+ topics** into a semantically searchable corpus using **Cohere embed-english-v3.0** (1024-dim embeddings) with automated rate-limit handling and retry logic.
 - Enforced **fully structured LLM outputs** across all 5 agent nodes using Pydantic v2 — zero unstructured text in the pipeline — with Groq (Llama 3.3 70B) as the reasoning engine delivering sub-second inference for real-time investigation streaming.
- Failure-Aware Vision – ML Perception Safety Engine | FastAPI, ROS2, OpenCV, WebSockets

Feb 2026

 - Achieved **zero false-safe resumptions** across 5 adversarial input types (blur, brightness, entropy, freeze, noise) via a trust-based safety state machine with deterministic policy gating and asymmetric temporal decay.
 - Sustained **30Hz real-time anomaly scoring** under adversarial conditions using a multi-signal frame reliability model with zero latency regression across 100+ simulated failure injections.
 - Cut debugging cycle time **60%** by migrating ROS2 prototype to FastAPI + WebSocket architecture with live telemetry streaming, model serving endpoints, and automated CSV diagnostics export.
- Intentra – GenAI Prompt Optimization Platform | TypeScript, LLM Orchestration, Cloudflare Workers

Nov 2025

 - Improved LLM output reliability by **67%** via a multi-stage prompt engineering pipeline — input scoring, semantic restructuring, and hallucination-risk validation — before model submission, targeting ambiguous and adversarial inputs.
 - Engineered quota-aware failover between **Gemini 1.5** and **Groq Llama 3.1** achieving **99.9% uptime** at **sub-100ms latency**; deployed on Cloudflare Workers edge with zero cold-start overhead and session-level rate limiting.

EDUCATION

- Bachelor of Technology in Computer Science Engineering

Expected May 2026

Lovely Professional University | CGPA: 9.08 / 10.0 (Top 5% of cohort)

Punjab, India

 - Coursework:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms, Statistical Analysis, Operating Systems, Computer Networks

CERTIFICATIONS

Generative AI with Large Language Models – DeepLearning.AI / Coursera | Algorithms Specialization – Stanford University